TP2 MASTER IOT

1. Context

Many connected objects use RF to communicate. These communications can be acquired by gateways built around SDRs (software defined radio).

Software-defined radio (SDR) is a radio communication system where components that conventionally have been implemented in analog hardware (e.g. mixers, filters, amplifiers, modulators/demodulators, etc.) **are implemented by means of software on a computer or embedded system.**

Superheterodyne receivers use a VFO (variable-frequency oscillator), mixer, and filter to tune the desired signal to a common IF (intermediate frequency) or baseband. Typically in SDR, this signal is then sampled by the analog-to-digital converter.

Software defined radios are of great use to military and mobile phone services, both of which must serve a wide variety of changing radio protocols in real time. In the long term, software-defined radios are expected to become the dominant technology in radio communications.

To acquire a wide frequency band, an analog system called a "superheterodyne receiver" is used. It allows the carrier frequency to be translated into baseband (the carrier is removed, all that remains is the modulated data)

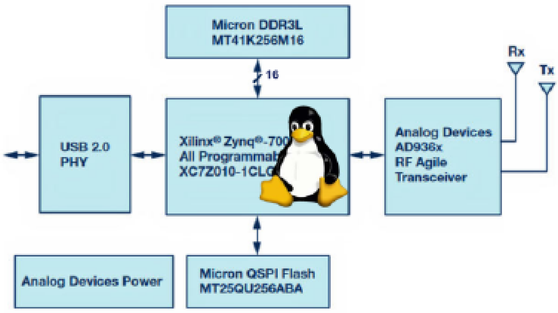
A superheterodyne receiver, is a type of radio receiver that uses frequency mixing to convert a received signal to a fixed intermediate frequency (IF) which can be more conveniently processed than the original carrier frequency.

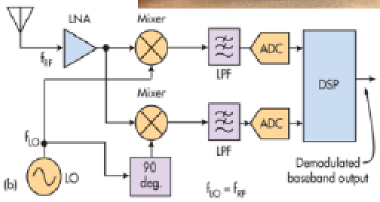
This technique is used in the Adalm pluto device.

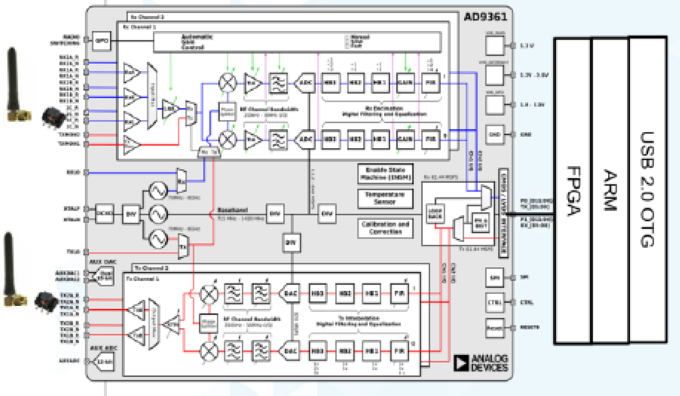
2. Presentation of the ADALM Pluto device

The PlutoSDR features independent receive and transmit channels that can be operated in full duplex. The PlutoSDR can generate or acquire RF analog signals from 325 MHz to 3800 MHz at up to 61.44 MegaSamples per second (MSPS)

- ADI Adalm-Pluto :





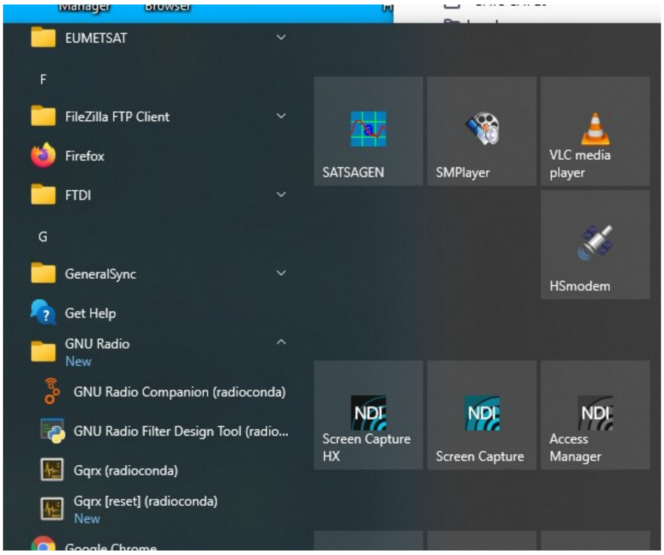


- Tuning range: 325-3800 MHz (expandable to 70-6000 MHz), 1 TX, 1 RX
- IQ samples capture: 12 bits, 65 kSPS to 61.44 MSPS, 200 kHz to 20 MHz signal bandwidth
- USB 2.0 interface
- Price : 230$ (215€)

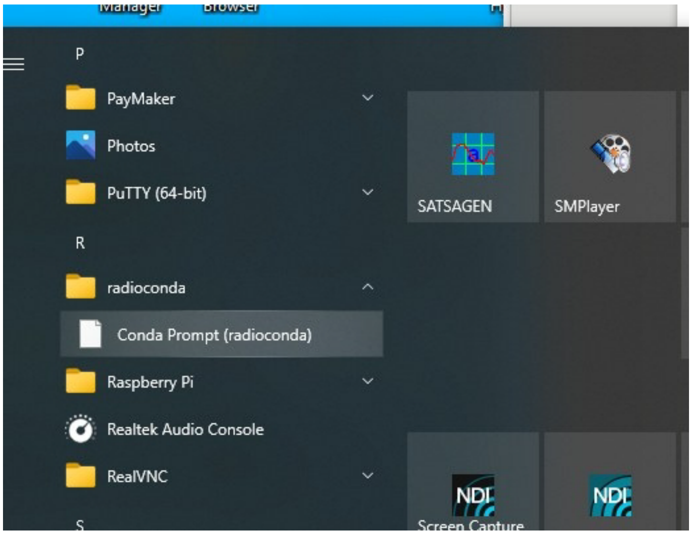
3. Checking SDR board driver installation

In the Windows Start Menu, there are two new directories called:

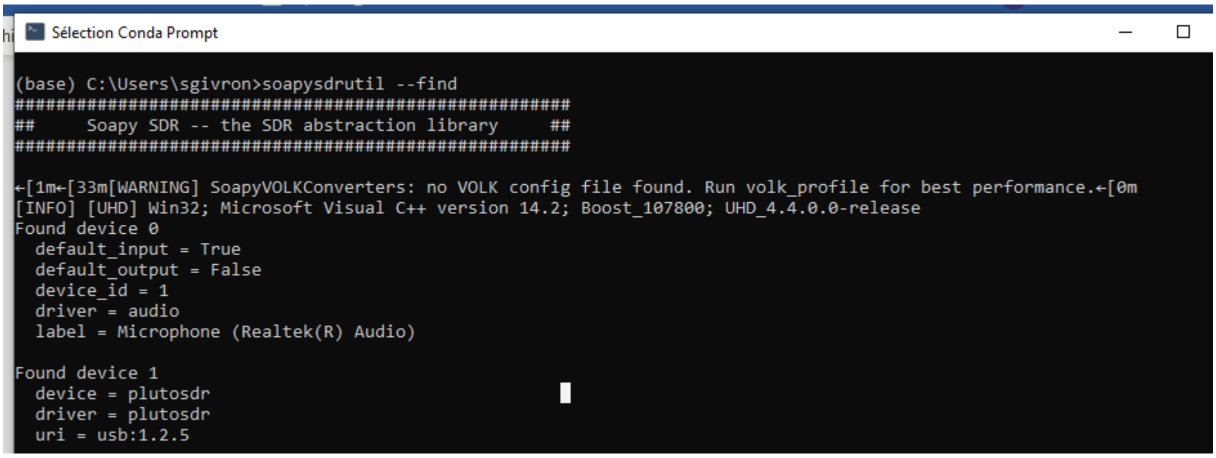
one GNU Radio



the other: **radioconda**



To check if the drivers are present for the Adalm-Pluto, you must run Conda Prompt
(radioconda).
execute the following command:
soapysdrutil --find
The result is, if the drivers are installed:




```
(base) C:\Users\sgivron>soapysdrutil --find
######################################################
##     Soapy SDR -- the SDR abstraction library     ##
######################################################

←[1m←[33m[WARNING] SoapyVOLKConverters: no VOLK config file found. Run volk_profile for best performance.←[0m
[INFO] [UHD] Win32; Microsoft Visual C++ version 14.2; Boost_107800; UHD_4.4.0.0-release
Found device 0
  default_input = True
  default_output = False
  device_id = 1
  driver = audio
  label = Microphone (Realtek(R) Audio)

Found device 1
  device = plutosdr
  driver = plutosdr
  uri = usb:1.2.5
```

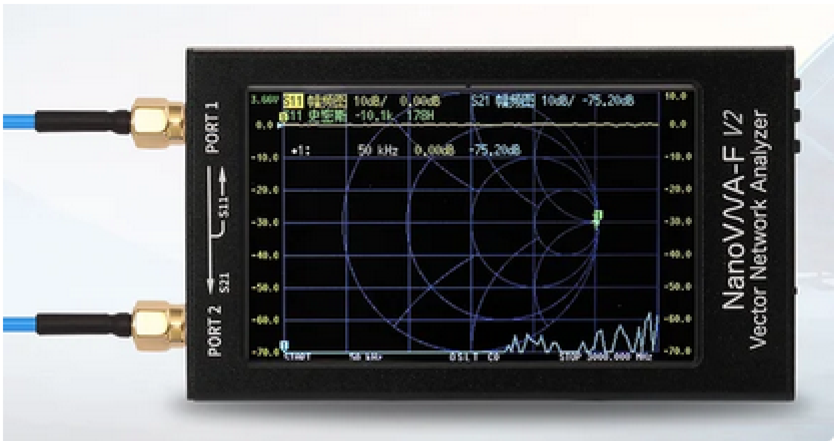
Experimental work:

You will work on the acquisition of 4 different signals >2.4GHz wifi signal >Lora signal at 868MHz, widely used in the Internet of Things >FM radio signal between 88MHz and 108MHz >DIO 1.0 connected socket signal at 433.92MHz

4. Antenna study

The first thing to do is to check that the antennas are adapted to the frequencies of the 4 signals

Use the VNA (Vector Network Analyser).



a. Study of the ADALM Pluto antenna_

Connect the antenna of the ADALM Pluto box to port 1 of the VNA.

The antennas connectors on the ADALM-PLUTO are standard polarity SubMiniature version A (SMA) connectors. There are many ways to characterize an antenna, from gain (or loss), radiation pattern, beamwidth, polarization, SWR and impedance.

The actual antennas included with the ADALM-PLUTO are Jinchang Electron JCG401. While the JCG401 is specified for 824-894 and 1710-2170MHz, they do operate over a much wider range, but it should be understood that all antennas are filters. They will have some frequency selectivity (however, they are not brick wall filters). Just because something is specified for 824 MHz, does not mean that it will not pick up something at 600MHz, or even 87MHz

Using the stylus, go to the following menus to study the SWR characteristic. SWR is used as a measure of impedance matching of a load to the characteristic impedance of a transmission line carrying radio frequency (RF) signals.

An antenna must always have its SWR between 1 and 2 in order to acquire the signal without loss. Beyond this value this means that very little signal is acquired

>On the VNA , choose DISPLAY /FORMAT/ check SWR and BACK

>On the VNA, TRACE/ select only the trace corresponding to swr. Then BACK 2 times to return to the main menu

In order to be able to move around the SWR characteristic, you will display a frequency marker.

>MARKER/SET FREQ   enter value 2.4G with the stylus

> Determine the frequency range in order to have a SWR between 1 and 2

b. <u>Study antenna n° 2</u>

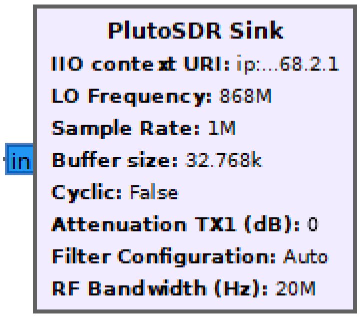
> Determine the frequency range in order to have a Swr between 1 and 2

Conclusion :

You have a first antenna well adapted to specific applications which will filter other unwanted frequencies (absence of disturbances from neighboring frequencies)

The second antenna can acquire a very wide band but may be disturbed by neighboring frequencies

5. Explanation of the transmission and reception blocks



IIO context URI

IP address of the unit, e.g. "ip:192.168.2.1" (without the quotes)

LO Frequency

Selects the TX local oscillator frequency.

Sample Rate

Sample rate in samples per second, this will define how much bandwidth your SDR transmits (the RF bandwidth parameter below just defines the filter). limits: >= 520833 and <= 61440000. A FIR (Finite Impulse Response) filter needs to be loaded or set to auto for values below 2.083 MSPS.

RF Bandwidth

Configures TX analog filters: TX BB LPF and TX Secondary LPF. limits: >= 200000 and <= 52000000

Buffer size

Size of the internal buffer in samples. The IIO blocks will only input/output one buffer of samples at a time. To get the highest continuous sample rate, try using a number in the millions.

Cyclic

Set to “true” if the “cyclic” mode is desired. In this case, the first buffer of samples will be repeated on the PlutoSDR until the program is stopped. The PlutoSDR IIO block will report its processing as complete: the blocks connected to the PlutoSDR IIO block won't execute anymore, but the rest of the flow graph will.

Attenuation TX1 (dB)

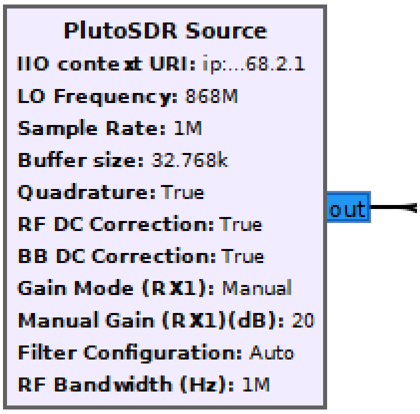
Controls attenuation for TX1. The range is from 0 to 89.75 dB in 0.25dB steps. Note: Maximum output occurs at 0 attenuation.

Filter

Allows a FIR filter configuration to be loaded from a file.

Filter Auto

When enabled loads a default filter and thereby enables lower sampling / baseband rates.



LO Frequency

Selects the RX local oscillator frequency.

IIO context URI

IP address of the unit, e.g. "ip:192.168.2.1" (without the quotes)

Sample Rate

Sample rate in samples per second, this will define how much bandwidth your SDR receives at once (the RF bandwidth parameter below just defines the filter). limits: >= 520833 and <= 61440000

RF Bandwidth

Configures RX analog filters: RX TIA LPF and RX BB LPF. limits: >= 200000 and <= 52000000

Buffer size

Size of the internal buffer in samples. The IIO blocks will only input/output one buffer of samples at a time. To get the highest continuous sample rate, try using a number in the millions.

Gain Mode (Rx1)

Selects one of the available modes: manual, slow_attack, hybrid and fast_attack. For most spectrum sensing type applications, use a manual gain, so that you actually know when a signal is present or not, and it's relative power.

Manual Gain (Rx1)(dB)

gain value, max of 71 dB, or 62 dB when over 4 GHz center freq

Filter

Allows a FIR filter configuration to be loaded from a file.

Filter Auto

When enabled loads a default filter and thereby enables lower sampling / baseband rates.

6. Frequency calibration 2.4GHz

An acquisition card must always be calibrated in frequency to ensure that the calibration of the card is correct.

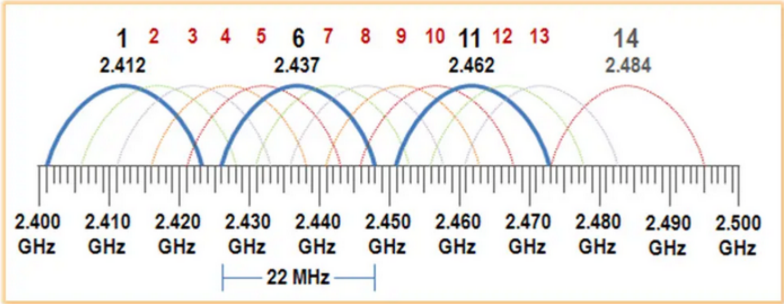
a. Connect Tx(VNA) to Rx (pluto) with SMA Male To SMA Male cable.

b. Use the **TP2_acqui_wifi.grc** file.
c. Set the VNA to have a carrier at 2.4MHz (touch the screen with the stylus, choose STIMULUS=>SIGNAL GENERATOR=> choose -9dB =>freq (set the frequency to 2.4GHz)=>check RF_OUT to have a signal in output of the VNA
d. Note any frequency deviation. Place this value in the frequency offset block and redo an acquisition to validate this correction.

7. Wifi Acquisition

On the 2.4 GHz frequency it is recommended to separate the bands by at least 4 channels, which corresponds to a bandwidth of 22 MHz. This will help you avoid any potential interference concerns.

>Using the **TP2_acqui_wifi.grc** file, adjust the freq and samp_rate variables to view channel 1 and channel 6 at the same time.



8. Module Lora

The transceiver LoRa® (long range) modem provides ultra-long range spread spectrum communication and high interference immunity whilst minimizing current consumption.

Lora was designed to link connected objects that are isolated or separated from the network infrastructure by several kilometers (1 to 40km). The transmitted signal is easy to interpret upon reception so that it can cover large distances.

The transmission frequency of the module is 868MHz.

9. Lora Frequency Calibration.

a. Connect Tx(VNA) to Rx (pluto).
b. Use the ***TP2_acqui_lora.grc*** file***.***
c. Set the VNA to have a carrier at 868MHz MHz (touch the screen with the stylus, choose STIMULUS=>SIGNAL GENERATOR=> choose -9dB =>freq (set the frequency to 868MHz)=>check RF_OUT to have a signal output of the VNA.
d. Note the frequency difference. Place it in the frequency offset block.

10. Lora frequency measurements.

a. Use the ***TP2_acqui_lora.grc*** file.
b. What is the carrier frequency of the Lora signal ?
c. Measure the bandwidth of the Lora signal.

11. Lora temporal measurements.

a. Use the **TP2_lora_demodulation.grc** file.
b. Measure the transmission duration of a frequency modulated frame.

12. FM radio signal.

The FM radio signal is a signal between 88MHz and 108MHz.

13. FM signal frequency calibration.

a. Connect Tx(VNA) to Rx (pluto).
b. Use the ***TP2_acqui_FM.grc*** file***.***
c. Set the VNA to have a carrier at 100MHz (touch the screen with the stylus, choose STIMULUS=>SIGNAL GENERATOR=> choose -9dB =>freq (set the frequency to 100MHz)=>check RF_OUT to have a signal output from the VNA.
d. Note the frequency difference. Place it in the frequency offset block.

14. Acquisition spectre signal FM

a. Use the ***TP2_acqui_FM.grc*** file
b. >Measure the bandwidth of an FM signal at f=107.9MHz.

15. FM signal spectrum demodulation.

a. Use the ***TP2_FM_sound_card.grc*** file.
b. Listen to the sound produced by the demodulated FM signal at f=107.9MHz.
c. Explain why you needed to decimate the flow.

16. Outlet DIO 1.0 home connected

This socket is controlled with 433.92MHz waves.

17. Frequency calibration .

a. Connect Tx(VNA) to Rx (pluto).
b. Use the ***TP2_acqui_DIO.grc*** file ***.***
c. Set the VNA to have a carrier at 433.92MHz MHz (touch the screen with the stylus, choose STIMULUS=>SIGNAL GENERATOR=> choose -9dB =>freq (set the frequency to 433.92MHz)=>check RF_OUT to have a signal at the output of the VNA.
d. Note the frequency difference. Place it in the frequency offset block.

18. Emission frequency search DIO 1.0 connected home

DIO connected home is a remote control for electrical outlets. The DiO 1.0 socket allows you to wirelessly control your lamps and electrical appliances.

a. Use the ***TP2_acqui_DIO.grc*** file.
b. Make the acquisition. Each group must purchase one order only (4 On and 4 Off, i.e. 8 orders)
c. Note the transmission frequency. Is there a frequency gap? Why ?

19. Saving the signal to a file.

a. Use the ***TP2_acqui_DIO.grc*** file.
b. Set enable, the file sink block. Indicate the path to the file storing the data.
c. Record an ON or OFF sequence of a take.

20. File Playback.

a. Use the ***TP2_play_DIO.grc*** file
b. Adjust the transmission frequency so that it is identical to the frequency of the remote control
c. Test that the digital sequence used to control the DIO

21. Decoding the ASK frame (OOK)

a. Use the ***TP2_aqui_DIO_trame.grc*** file
b. How long does a frame last?
c. What is the type of modulation?